Proficient in AutoCAD (Civil) (80 Hours)				
S.No	**Session**	**Topics Covered**	**Type**	**Duration (hrs)**
1	Session 1	Introduction to engineering Drawings and AutoCAD Introduction to Engineering Drawings Various types of engineering drawings used in AEC industry Introduction to AutoCAD Exploring GUI and Workspaces Co-ordinate systems **Display Control :** Zoom, Pan, Redraw, Regen, Clean Screen, Steering wheels **File Management:** New, Qnew, Open, Save, Save as, Close, Exit, Quit	Theory/Demo	2.0 Hrs
2	Session 2	Practice 01: Exercises included for Practice	Practical	2.0 Hrs
3	Session 3	Introduction to engineering Drawings and AutoCAD **Tutorial 1:** Working with Co-ordinate systems **Tutorial 2:** Draw using Absolute Co-ordinate system **Tutorial 3:** Draw using Relative Co-ordinate system **Tutorial 4:** Draw using Polar Co-ordinate system	Theory/Demo	2.0 Hrs
4	Session 4	Practice02: Exercises included for Practice	Practical	2.0 Hrs
5	Session 5	Drafting Basic Geometry Shapes in AutoCAD **Drawing settings:** Units, Limits Drafting settings Snap and Grid Polar tracking Object snap 3D Object snap Dynamic input (F1 to F12 Function keys) **Drawing tools:** Line, Circle, Arc, Ellipse, Donut, Polygon, Rectangle, Point, Multiple point	Theory/Demo	2.0 Hrs
6	Session 6	Practice 03: Exercises included for Practice	Practical	2.0 Hrs
7	Session 7	Drafting Basic Geometry Shapes in AutoCAD **Modify tools :** Erase, Undo, Redo, Explode, Move, Copy, Rotate, Mirror, Array, Align, Scale, Stretch, Lengthen, Trim, Extend, Break, Join, Chamfer, Fillet, Blend curves, Object selection **Object properties :** Properties window Color Linetype Linetype scale Line weight Match properties **Tutorial 5:** Road junction - Layout	Theory/Demo	2.0 Hrs
8	Session 8	Practice 04: Exercises included for Practice	Practical	2.0 Hrs
9	Session 9	Advanced Drafting and Modifying Tools in AutoCAD Drawing Tools Multiline Pline Spline Xline Ray Wipeout Revision cloud **Modify Tools :** Multiline style Multiline edit Polyline edit Spline edit, Offset PD Mode Grip editing PD Size Edit Array Change space Reverse Copy nested objects Delete duplicate objects	Theory/Demo	2.0 Hrs

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		Bring or send Objects		
		Views : Grip Editing		
10	Session 10	Practice 05: Exercises included for Practice	Practical	2.0 Hrs
11	Session 11	Advanced Drafting and Modifying Tools in AutoCAD **Tutorial 6 :** Landscape Design **Tutorial 7 :** Building Floor Plan Drafting	Theory/Demo	2.0 Hrs
12	Session 12	Practice 06: Exercises included for Practice	Practical	2.0 Hrs
13	Session 13	Layer Management Layer Management Adding / Removing Layers Layer Status New Property Filter New Group Filter Layer Status Manager Plot Control **Object Selection Methods :** Select, Qselect, Filter **Tutorial 8:** Creation of Layers with various Properties **Tutorial 9:** Interior Design using Single Point Perspective Method	Theory/Demo	2.0 Hrs
14	Session 14	Practice 07: Exercises included for Practice	Practical	2.0 Hrs
15	Session 15	Hatching Techniques **Hatching utilities:** Hatch Hatchedit Gradient Boundary Hatch Settings Fill and Fill mode **Inquiry commands:** Id, Dist, List, Radius, Angle, Area, Volume **Isometric drawings:** Isometric snap Isometric drawings **Tutorial 10:** Understand the Hatch and Gradient Concepts Beams of Different depths	Theory/Demo	2.0 Hrs
16	Session 16	Practice 08: Exercises included for Practice	Practical	2.0 Hrs
17	Session 17	Dimensioning, Tolerancing and Table Creation Techniques **Dimensions:** Linear, Aligned, Radius, Diameter, Angle, Arc length, Continuous, Baseline, Tolerance, Dimension Space, Dimension Break, Inspection, Jogged radius, Ordinate dimensions, Leader, Qleader, Oblique, Dimensions Text angle, Dimensions Text Alignment, Quick dimension Leader Multi leader Multi leader style Add leader Align Leader Collect Leader Remove Leader Annotation Tools Text Style Multiline Text Scale Text Spell Table Table style Table edit Dimension style Manager Dimension edit Dimension text edit Dimension Associative Re-associate Jog line Inspect Update	Theory/Demo	2.0 Hrs

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18	Session 18	Practice 09: Exercises included for Practice	Practical	2.0 Hrs
19	Session 19	Dimensioning, Tolerancing and Table Creation Techniques **Tutorial 11:** Dimensioning the Road junction-Layout in AutoCAD **Tutorial 12:** Structural Plate Detailing	Theory/Demo	2.0 Hrs
20	Session 20	Practice 10: Exercises included for Practice	Practical	2.0 Hrs
21	Session 21	Parametric Drawing Parametric Modeling Geometric Constraints Dimensional Constraints Delete constraints **Tutorial 13:** Applying Geometrical and Dimensional constraints	Theory/Demo	2.0 Hrs
22	Session 22	Practice 11: Exercises included for Practice	Practical	2.0 Hrs
23	Session 23	Application of Blocks and Attributes Create Block Base Write block Dynamic Block Insert Design Center Tool Palette Divide and Measure (Use block and point options) **Attributes:** Attribute definition, Attribute edit, Edit single & multiple attribute, Attribute Display, Manage Attributes, Attdia, Attext, Eattext **Standards :** Drawing sheets standards and Title block standards Title blocks creation, Symbol creation, BOM creation, Data extraction and Link Data **Tutorial 14:** Creating a Door Symbol using Block and Dynamic Block Methods **Tutorial 15:** Creating a Title Block for a Drawing Sheet using Attributes	Theory/Demo	2.0 Hrs
24	Session 24	Practice 12: Exercises included for Practice	Practical	2.0 Hrs
25	Session 25	**External References:** Xref, Xbind, Circular Xref, Refedit **External References:** Xref, Xbind, Circular Xref, Refedit Action Recorder **Express Tools:** Explode Attribute Arc Aligned Text Explode Text Case Torient Text Fit Text circle Modify text Super Hatch Make Line type, Shape creation **Tutorial 16:** Creating a Master Plan drawing using XREF	Theory/Demo	2.0 Hrs
26	Session 26	Practice 13: Exercises included for Practice	Practical	2.0 Hrs
27	Session 27	Object Linking and Embedding **OLE concepts:** Xref Xbind Circular Xref Refedit Copy Copybase Copylink Pasteclip Pastespecial Pasteblock Pasteoriginal **Tutorial 17:** Object Linking in AutoCAD	Theory/Demo	2.0 Hrs
28	Session 28	Practice 14: Exercises included for Practice	Practical	2.0 Hrs
		Layout, Plot and Publish Etransmit, Publish, Publish to web Layout, Page setup		

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29	Session 29	Viewports Viewport Lock and Unlock Viewport Layers Management Model space Paper space Plot styles and Plot (Print) in Model and Layout area **Tutorial 18:** Understanding Layouts, Viewport Management and Plotting	Theory/Demo	2.0 Hrs
30	Session 30	Practice 15: Exercises included for Practice	Practical	2.0 Hrs
31	Session 31	Basic 3D Modeling in AutoCAD Introduction to 3D Modeling **Basic Modeling Tools:** Primitives Extrude Sweep Revolve Loft Press pull Blend Patch Surface Offset Surface Trim Surface untrim	Theory/Demo	2.0 Hrs
32	Session 32	Practice 16: Exercises included for Practice	Practical	2.0 Hrs
33	Session 33	Basic 3D Modeling in AutoCAD **Basic Editing Tools:** Union, Subtract Intersect Slice Extrude Faces Shell Fillet Chamfer UCS Co-ordinates Viewport controls View controls Visual controls Section Import & Export options **Layout:** Base View from Model space and Inventor View creation Projected Section Detail views **Tutorial 19:** Creating a 3D Model in AutoCAD	Theory/Demo	2.0 Hrs
34	Session 34	Practice 17: Exercises included for Practice	Practical	2.0 Hrs
35	Session 35	Introduction to AutoCAD Architecture Introduction to AutoCAD Architecture Exploring the GUI of AutoCAD Architecture Tool palettes Properties window Content browser Style manager	Theory/Demo	2.0 Hrs
36	Session 36	Practice 18: Exercises included for Practice	Practical	2.0 Hrs
37	Session 37	Introduction to AutoCAD Architecture AEC Objects Grid Walls and curtain walls Doors and windows Roofs and slabs Stairs Spaces Sections and elevations	Theory/Demo	2.0 Hrs

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38	Session 38	Practice 19: Exercises included for Practice	Practical	2.0 Hrs
20	Session 20	Introduction to AutoCAD Architecture	Theory/Demo	2.0 Hrs
		Project browser		
		Constructs		
		Elements		
		Views		
		Tutorial 20: Architectural Drafting in AutoCAD Architecture		
40	Session 40	Practice 20: Exercises included for Practice	Practical	2.0 Hrs